

Rabin Khadka

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Aspiring doctoral researcher in data science and supply chain analytics, focused on explainable machine learning for demand forecasting, inventory decisions, and price transmission in supply chains.

Education

Master of Science in Data Science

2023 – 2025

Carolina University, Winston-Salem, North Carolina, USA — GPA 3.83 / 4.00

- Graduate training in statistical analysis, machine learning, predictive modeling, and quantitative methods.

Bachelor of Business Administration

Texas College of Management & IT, Kathmandu, Nepal (affiliated with Lincoln University College, Malaysia) — GPA 3.90 / 4.00

- Undergraduate training in business administration, quantitative methods, management, and organizational decision-making.

Research Interests

Supply Chain Analytics · Business Analytics · Machine Learning · Explainable AI · Demand Forecasting · Inventory Management · Procurement & Supplier Management · Predictive Modeling

Selected Research & Analytical Projects

Upstream Price Pressures and Consumer Inflation: Testing Supply Chain Price Transmission in the U.S. *(working paper)*

- Analyzed U.S. CPI, Producer Price Index, and Import Price Index data (BLS, Jan 2015–Jul 2026) using cross-correlation and pairwise Granger-causality F-tests.
- Found import prices Granger-cause CPI changes ($p = 0.021$, one-month lag) with no reverse causality — evidence of a one-directional, exogenous upstream shock.
- Tools: Python, NumPy, Pandas, SciPy, Matplotlib

Predicting Global Life Expectancy Using Machine Learning

- Compared Linear Regression, Decision Tree, and Random Forest models on WHO life-expectancy data (193 countries, 2000–2015).
- Random Forest achieved the strongest performance (test $R^2 = 0.967$); HIV/AIDS mortality and Income Composition of Resources accounted for ~77% of feature importance.
- Tools: Python, Pandas, Scikit-learn, Matplotlib, Seaborn

Amazon Sales & Customer Sentiment Analysis

- Modeled discount depth from price, rating, and category across 1,465 Amazon products; scored review sentiment with TextBlob and VADER.
- A tuned Gradient Boosting model raised test R^2 to 0.32, roughly double the linear baseline (0.16), indicating non-linear discounting behavior.
- Tools: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, TextBlob, NLTK

Insurance Fraud Detection Using Machine Learning

- Applied SMOTE for class-imbalance correction and compared Random Forest, Gradient Boosting, SVM, and neural-network classifiers on insurance claims data.
- Evaluated generalization across training, validation, and test splits, examining overfitting risk under severe class imbalance.
- Tools: Python, Pandas, Scikit-learn, SMOTE, TensorFlow/Keras, Matplotlib, Seaborn

Professional & Teaching Experience

Junior Social Media Data Analyst

Apr 2025 – Present

Kaebox

- Analyze social media performance and engagement data to identify trends and support data-driven marketing decisions.
- Monitor key performance indicators and evaluate digital content performance across platforms.

Business Organization Instructor

2020 – 2022

Wits Academy, Boudha, Kathmandu, Nepal

- Taught Business Organization to Grade 11 and 12 students, covering management, organizational structure, and decision-making.
- Guided students in developing analytical thinking and applying business concepts to practical situations.

Technical & Quantitative Skills

Programming: Python, R, SQL

Data Science: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Matplotlib, Seaborn

Methods: Machine learning, predictive modeling, statistical analysis, classification, regression, PCA, model evaluation, Granger causality